MACROGOL

Faecal impaction

Mix with water

CHEMICAL NAME Polyethylene glycol 3350 or 4000 ($HO(CH_2CH_2O)_nH$) ± electrolytes

CLASS Osmotic laxative (iso-osmotic)

TRADE NAMES Movicol · Laxido · MiraLAX · Forlax · Klean-Prep (bowel prep)

THERAPEUTIC USES / INDICATIONS

- Chronic constipation (adults and children; first-line in children).
- Faecal impaction (high-dose regimen).
- Bowel cleansing before colonoscopy (large-volume preparations).

PHARMACODYNAMICS

- Large polymer holds water in colon by osmosis → ↑ stool volume and softness; electrolytes prevent net fluid/electrolyte loss.

ADMINISTRATION

- **PO**: dissolve each sachet in 125 mL water (paediatric sachets 62.5 mL).
- Can be mixed with juice; drink within 6 h of mixing.
- Bowel prep: specific large volumes per protocol.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Abdominal distension, bloating, borborygmi.
- Nausea.
- Diarrhoea.

ADVERSE EFFECTS (SERIOUS)

- Electrolyte disturbance, dehydration (large volumes, elderly).
- Allergic reactions (rare).

DOSAGES

- **Constipation (adult):** 1–3 sachets daily (13.8 g each).
- **Faecal impaction:** 8 sachets/day (dissolved in 1 L) over 6 h for up to 3 days.
- **Children (paediatric sachet 6.9 g):** <1 yr ½–1 daily; 1–6 yr 1–4 daily; impaction escalating regimen.

CONTRAINDICATIONS & CAUTIONS

- Intestinal obstruction or perforation; severe inflammatory bowel disease, toxic megacolon.
- Ileus.

KEY INTERACTIONS

- May ↓ absorption of other oral drugs taken at the same time (especially during bowel prep).

PHARMACOKINETICS

- **Onset:** 1–3 days (constipation); impaction clears in 1–3 days.
- Not absorbed or metabolised.
- Excreted in faeces.

NURSING MANAGEMENT

- Make sure correct dilution.
- Monitor bowel movements, abdominal distension, fluid balance.
- Elderly: monitor electrolytes with large volumes.

PATIENT EDUCATION

- Mix each sachet with the right amount of water.
- Expect a soft motion within 1–3 days.
- Children: continue maintenance for months as advised.

KEY POINTS

- Osmotic iso-osmotic laxative.
- Disimpaction regimen up to 8 sachets/day.
- First-line in children's constipation.
- Dissolve in 125 mL water.